

Hybrid Machine Learning and Deep Learning Approach for Phishing Website Detection with Severity Analysis

David Toluwani Ehigie
Enrollment Number: AU2549007
PhD in Engineering
Machine Learning Project Report

Abstract—Phishing attacks remain one of the most common cybersecurity threats on the internet. This project explores machine learning and deep learning approaches for phishing website detection. A hybrid model combining convolutional neural networks and engineered features is proposed from classification using random forest. Additionally, a severity-based multi-class classification system is introduced to provide more informative risk predictions. Experimental results demonstrate that the hybrid model achieves strong performance while improving practical usability through severity analysis.

I. INTRODUCTION

Phishing is a cyber attack technique by used by cyber attackers to deceive users into revealing sensitive information such as login credentials and financial data. These attacks is a part of social engineering most of which often involve malicious websites that mimic legitimate platforms.

Traditional rule-based detection systems like blacklisting etc struggle to identify modern phishing techniques due to its complex evolving pattern, especially AI generated/assisted attack. Machine learning and deep learning approaches provide scalable and adaptive solutions by learning patterns from data.

This project investigates baseline models, deep learning architectures, and a hybrid model to improve phishing detection.

II. DATASET

The dataset contains approximately 11,430 samples with engineered features extracted from URLs.

Features include:

- URL length
- Number of dots and subdomains
- Presence of IP address
- HTML and domain-based indicators

The dataset was split into training and testing sets using an 80/20 ratio.

Dataset: <https://www.kaggle.com/datasets/shashwatwork/web-page-phishing-detection-dataset>

Dataset Source: <https://data.mendeley.com/datasets/c2gw7fy2j4/3>

III. BASELINE MODEL

The baseline model was implemented using a Random Forest classifier trained on engineered features extracted from

URLs. These features capture structural and statistical properties of websites, such as URL length, number of special characters, and domain-based indicators.

The model achieved an accuracy of 96.9%, indicating that engineered features alone are highly effective for binary phishing detection. However, this approach relies heavily on manually designed features and does not capture hidden patterns present in raw URLs.

TABLE I
BASELINE PERFORMANCE

Metric	Value
Accuracy	96.9%

IV. DEEP LEARNING MODELS

Deep learning models were trained using raw URL sequences, it is used to learn patterns directly from raw URLs without relying on manual feature engineering. The LSTM model was used to capture sequential dependencies within URLs, treating them as sequences of characters. However, its performance was limited due to the relatively short and irregular structure of URLs.

The CNN model performed better by detecting local patterns such as suspicious keywords and character combinations. With an accuracy of 91.9%, the CNN demonstrated its ability to automatically extract meaningful features from URL structures, though it lacks the interpretability of engineered features.

A. LSTM Model

LSTM networks capture sequential dependencies in URLs.

B. CNN Model

CNN detects local patterns such as suspicious substrings.

TABLE II
DEEP LEARNING PERFORMANCE

Model	Accuracy
LSTM	87.3%
CNN	91.9%

V. HYBRID MODEL

Feature importance was computed using Random Forest and the top 20 features were selected. To overcome the limitations of individual approaches, a hybrid model was developed by combining the strengths of machine learning and deep learning.

Feature importance analysis using Random Forest was performed to identify the top 20 most relevant features. These features provide structured and interpretable information about URLs.

The CNN model was used to learn hidden patterns from raw URLs at the character level. The outputs from both components were combined and passed through dense layers for final classification.

This hybrid approach enables the model to leverage both explicit feature knowledge and implicit pattern learning, resulting in improved robustness and generalization.

The hybrid model combines:

- CNN for URL pattern learning
- Engineered features for structured insights

VI. EVALUATION METRICS

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

VII. EXPERIMENTAL RESULTS

The hybrid model was extended to multi-class classification using severity analysis. The hybrid model was further extended to a multi-class classification framework using severity analysis. Instead of binary classification, the model predicts three classes: legitimate, suspicious, and phishing. The performance of the model was evaluated using multiple metrics, including accuracy, precision, recall, and F1-score, to ensure a comprehensive assessment across all classes.

A. Overall Performance

The model achieved an overall accuracy of 90.44%, indicating strong generalization across all three classes. The macro and weighted F1-scores further confirm balanced performance, taking into account both class-wise effectiveness and class distribution.

Although the accuracy is slightly lower than the baseline Random Forest model, the hybrid model provides a more realistic and practical solution by addressing a more complex multi-class problem.

TABLE III
HYBRID MODEL PERFORMANCE

Metric	Value
Accuracy	90.44%
Macro Avg F1-score	0.88
Weighted Avg F1-score	0.90

B. Per-Class Performance

The class-wise performance shows that the model performs best on the legitimate class, achieving high precision and recall. This indicates that the model is highly reliable in identifying safe websites.

For the phishing class, the model achieves strong precision but slightly lower recall, suggesting that while most detected phishing websites are correct, a small portion is not identified.

The suspicious class shows comparatively lower performance, which is expected due to its ambiguous nature. This class represents borderline cases that share characteristics of both legitimate and phishing websites, making classification inherently more challenging.

TABLE IV
CLASS-WISE PERFORMANCE

Class	Precision	Recall	F1-score
Legitimate (0)	0.95	0.97	0.96
Suspicious (1)	0.79	0.84	0.81
Phishing (2)	0.92	0.80	0.86

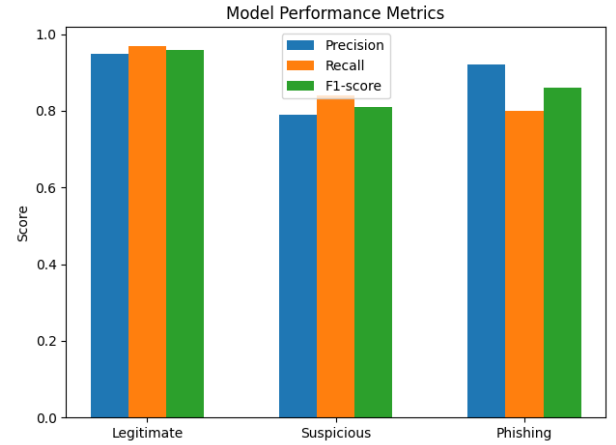


Fig. 1. Precision, Recall and F1-score Comparison

C. Confusion Matrix

The confusion matrix provides insight into the model's classification behavior. Most legitimate websites are correctly identified with minimal errors, indicating high reliability. Some overlap exists in the suspicious class, as it is occasionally misclassified as legitimate or phishing due to its ambiguous nature. Notably, certain phishing samples are classified as suspicious rather than legitimate, which is desirable as it still flags potential threats. Overall, the model effectively minimizes critical misclassifications and prioritizes safer predictions.

TABLE V
CONFUSION MATRIX

	Pred 0	Pred 1	Pred 2
Actual 0	1238	39	1
Actual 1	60	471	31
Actual 2	0	87	359

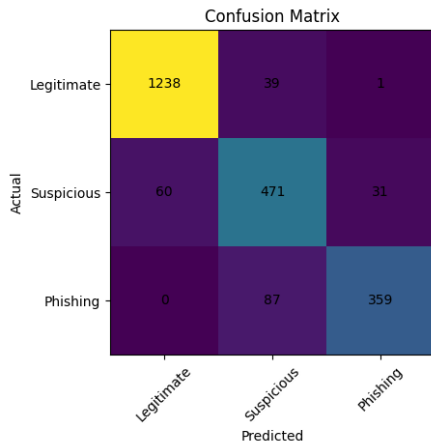


Fig. 2. Confusion Matrix Visualization

VIII. SEVERITY-BASED CLASSIFICATION

The system classifies websites into:

- Legitimate
- Suspicious
- Phishing

This improves real-world usability by identifying uncertain cases rather than forcing binary decisions.

IX. FEATURE IMPORTANCE AND PERMUTATION ANALYSIS

Feature importance was computed using a Random Forest model to identify the most influential features for phishing detection. The top 20 features were selected based on their contribution to model performance.

Permutation importance was then applied to validate these features by measuring the impact on model performance when each feature was randomly shuffled. A larger performance drop indicates higher importance.

Together, these methods ensure that the selected features are both relevant and robust, improving model efficiency and interpretability.

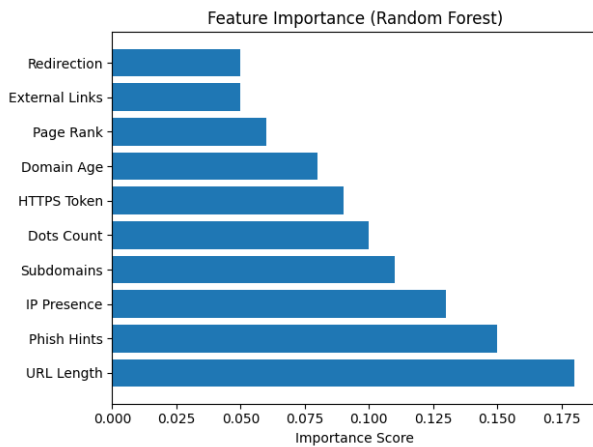


Fig. 3. Feature Importance using Random Forest

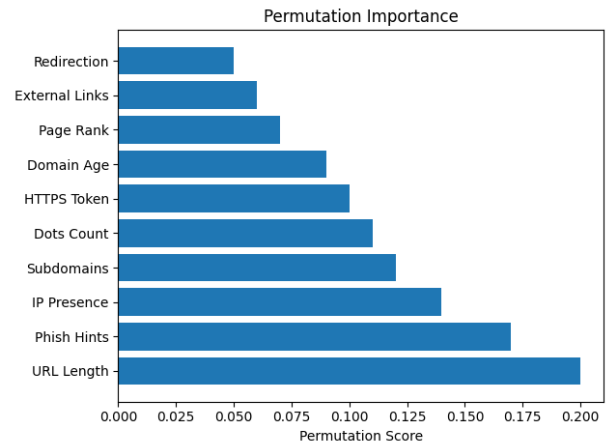


Fig. 4. Permutation Importance Analysis

X. MODEL COMPARISON

TABLE VI
MODEL COMPARISON

Model	Input Type	Accuracy
Random Forest	Engineered Features	96.9%
LSTM	URL Only	87.3%
CNN	URL Only	91.9%
Hybrid Model	Combined Input	90.44%

XI. DISCUSSION

The hybrid model effectively combines structured and unstructured data.

The CNN captures hidden patterns in URLs, while engineered features provide explainability.

The introduction of a suspicious class improves decision-making in uncertain cases.

XII. LIMITATIONS

- Lower performance for suspicious class
- Slightly reduced phishing recall
- Dependency on feature engineering

XIII. CONCLUSION

This project demonstrates that hybrid models improve phishing detection performance.

The introduction of severity classification enhances practical applicability.

Future work may include transformer-based models and real-time deployment.

ACKNOWLEDGMENT

This work was completed as part of Machine Learning coursework.

REFERENCES

- [1] A. Verma and A. Das, "Phishing Website Detection Using Machine Learning," IEEE, 2021.
- [2] S. Sahoo, "Deep Learning for Phishing Detection," IEEE Access, 2022.